

CROSS-LINGUAL ACTIVATION STEERING IN LARGE
LANGUAGE MODELS: REPRESENTATIONAL ALIGNMENT AND
BEHAVIOURAL TRANSFER

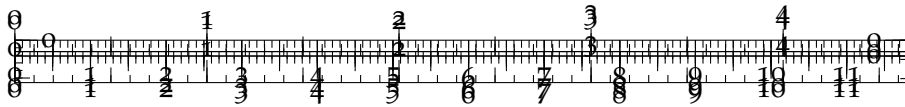
MISUN KIM

MSc Dissertation

MSc Speech and Natural Language Processing
School of Computer Science
Faculty of Engineering
University of Sheffield

Supervisors: Dr Cass Zhao

September 2026

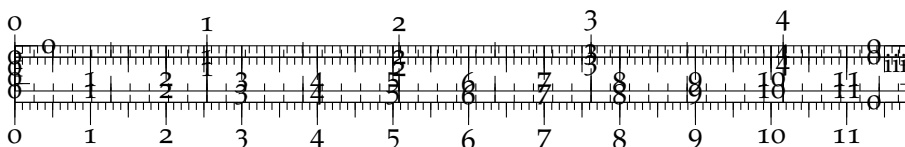


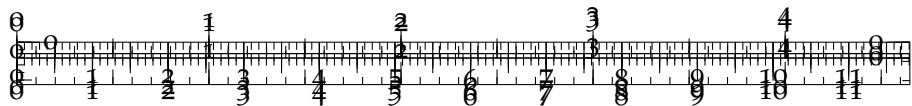
ABSTRACT

Large language model alignment methods are often developed and evaluated in English, but their behaviour may not transfer consistently across languages. This dissertation investigates the cross-lingual generalisation of Contrastive Activation Addition (CAA) for reducing sycophantic behaviour in large language models. We construct and apply steering directions across three open-weight models and five languages: English, Italian, Chinese, Korean, and Kazakh. We evaluate the interventions using held-out test sets, layer-wise analysis, bootstrap confidence intervals, and random and permutation controls. We additionally construct separate English-derived truthfulness steering vectors for a forced-choice TruthfulQA task to examine whether cross-lingual transfer is behaviour-dependent.

Our experiments show that cross-lingual transfer is model- and behaviour-dependent. English-derived sycophancy steering transferred reliably across all five languages in Qwen3.5-9B and Gemma4-12B-IT, but not in OLMo3-7B-Instruct, where only the English effect was statistically reliable. A cross-task evaluation revealed a dissociation: Qwen3.5-9B, which transferred sycophancy steering reliably, showed no consistent pattern of cross-lingual truthfulness transfer at the English-selected layer, whereas OLMo3-7B-Instruct, which failed to transfer sycophancy, produced reliable truthfulness gains in three of four non-English languages.

Across models and behaviours, strongly aligned steering directions produced uneven behavioural transfer, and in OLMo3-7B-Instruct no non-English sycophancy effect was statistically reliable. High cross-lingual representational similarity is therefore not sufficient to guarantee comparable behavioural effects. Whether such similarity is necessary remains unresolved. These results indicate that the cross-lingual reliability of alignment interventions should be evaluated through behavioural outcomes in each target language, rather than inferred from representation geometry alone.





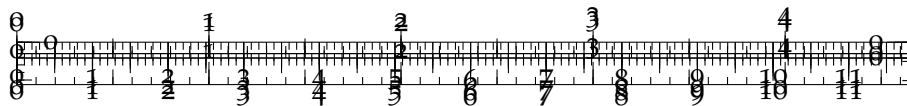
ACKNOWLEDGMENTS

I would like to thank my supervisor, Dr Cass Zhao, for her guidance, encouragement, and thoughtful feedback throughout this dissertation. I am also grateful to Dr Marco Valentino for his support and valuable advice.

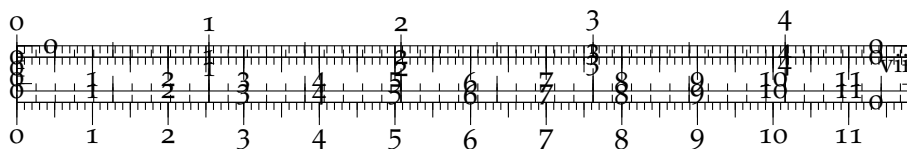
My thanks also go to Ali for his help with multilingual translations and data preparation, and to Feng for his helpful feedback on the project.

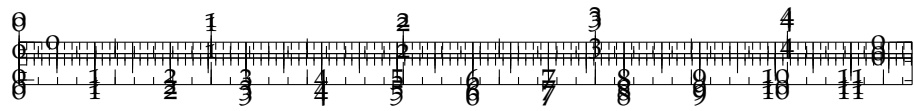
Finally, I would like to thank my family and friends for their continued support throughout my MSc.





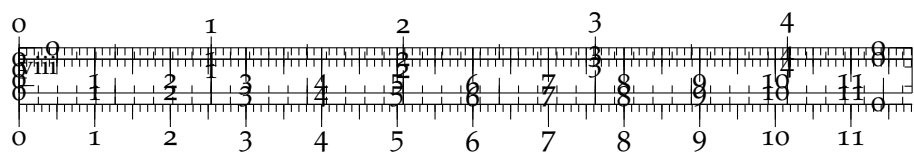
CONTENTS





LIST OF FIGURES

LIST OF TABLES



DECLARATION

I declare that this dissertation is my own work and that all sources used have been appropriately acknowledged and referenced. This work has not been submitted, in whole or in substantial part, for any other degree or qualification.

Sheffield, United Kingdom, September 2026

Misun Kim